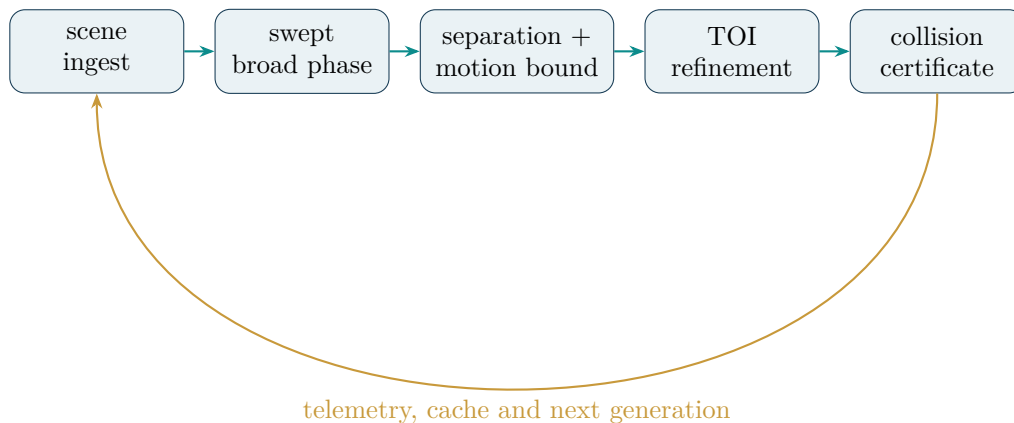


Non-Euclidean Holographic Data Fields

CCD-Centered Formalization

Cradle-to-Grave Validation Dossier, Version 0.4



Release position

This document turns the NHDF roadmap toward one falsifiable center: general-purpose continuous collision detection as an architecture, with an executable primitive reference and explicit pathways to robust mesh, convex, SDF, CPU, GPU, and application implementations. Optical, quantum, chemical, biological, medical, SAR, fabrication, and cosmological material is retained only through separate adapter contracts and evidence gates.

Tom Klootwijk

1 September 2026

Document status: research specification and executable validation scaffold. Not safety-certified, medically validated, cryptographically authoritative, or a claim that physical reality is an NHDF substrate.

Document control

Field	Value
Title	NHDF-CCD: CCD-Centered Formalization and Cradle-to-Grave Validation Dossier
Version	0.4.0
Author attribution	Tom Klootwijk
Date	1 September 2026
Primary center	Continuous Collision Detection (CCD)
Executable status	Python CPU reference; C++17 exact-sphere slice; CUDA source skeleton
Bundled verification	22 Python tests, one C++ smoke test, four reference vectors, 5,000-query generated differential benchmark
Evidence class	Conceptual architecture plus bounded executable scaffold; no production or safety certification
Privacy rule	Civil identifiers and date of birth are not included in executable state or used as cryptographic material

Abstract

The supplied NHDF work develops an ordered vocabulary of log-polar addressing, local zero-level constraints, parity events, bounded branching, forward kinematics, analytic sweeps, projection, and feedback. The earlier formal specification made the essential non-degenerate correction that a global field satisfying $F \equiv 0$ is unusable and that each valid cell must instead lie on a nontrivial local constraint. The commute transcript then connected an SDF lookup and kinematic motion to Continuous Collision Detection, but overstated that connection as a complete one-clock solution. The whole-substrate extension mapped the vocabulary into sensors, actuators, chemistry, biology, medicine, photonics, quantum optics, fabrication, SAR, and cosmology.

Version 0.4 centers the research program on CCD because it supplies a precise mathematical question, exact and conservative reference methods, public benchmark datasets, explicit failure states, and a direct route from CPU semantics to GPU and application integration. A pairwise separation function $d_{ij}(t)$ defines the contact zero set. Exact primitive solvers, conservative advancement, interval inclusion, swept broad-phase bounds, and bounded queues produce a certificate with one of seven states: `HIT`, `NO_HIT`, `INITIAL_OVERLAP`, `INCONCLUSIVE`, `UNSUPPORTED`, `INVALID_INPUT`, or `CAPACITY_EXCEEDED`. The NHDF log-polar, parity, golden-ratio, and topology constructs remain traceable as scheduling and experimental operators, but none may certify or suppress a collision unless an independent equivalence proof is established.

The accompanying package implements exact linear sphere–sphere, sphere–plane, and swept-AABB queries; conditional conservative advancement for declared separation oracles; point/sphere versus a true or conservatively bounded SDF under translation; a deterministic broad phase; interval-refinement ablations; schemas; telemetry; tests; benchmarks; a C++ slice; and a GPU plan. The remaining substrate materials are mapped through a uniform adapter contract that preserves domain-specific state, evolution laws, units, invariants, uncertainty, validation metrics, and stop conditions.

Executive determination

What this release establishes

NHDF can be developed as a bounded, causal operator architecture around CCD. The scientifically defensible nucleus is the ordered closure of local constraints, kinematic prediction, conservative swept bounds, bounded refinement, observable certificates, and feedback telemetry. Version 0.4 makes that nucleus executable for a deliberately small primitive profile and gives every later implementation a testable contract.

What this release does not establish

The release does not solve every possible CCD query, does not implement triangle-mesh or rotating-body robustness, does not include contact response, and does not demonstrate a GPU or physical-substrate advantage. It does not validate medical treatment, biological Cross-Kerr computation, universal golden-ratio branching, storm immunity, zero-latency optics, or cosmological holography. Those claims are either rejected or retained as separately gated hypotheses.

Immediate next gate

The next scientific milestone is not a larger vocabulary. It is robust differential validation against exact time-of-impact corpora and independent implementations for vertex–face and edge–edge triangle queries, followed by CPU/GPU equivalence. Only after that gate should an application claim be made.

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Part I

Evidence boundary and architectural turn

Purpose, scope, and release philosophy

1.1 Why CCD becomes the center

A useful research center must expose a clear input, a clear output, a correctness criterion, failure conditions, baselines, and a way to reproduce results. CCD satisfies all six. Given continuous object trajectories over a time interval, the engine must determine whether the objects first touch, identify or bound the earliest time, and avoid the tunneling failure of endpoint-only testing. Robust CCD research also provides exact or analytic datasets, inclusion-based algorithms, open implementations, and known degeneracies against which a new architecture can be tested (B. Wang, Ferguson, Schneider, et al., 2021; Belgrod et al., 2023).

The turn toward CCD does not discard the earlier NHDF work. It changes the burden of proof. Terms such as “local zero set,” “forward timeline,” “bounded branch tree,” and “analytic sweep” receive direct geometric meanings. Terms such as parity, golden-ratio scaling, and non-orientable chart flips are kept as optional scheduling or diagnostic variables until an ablation establishes their value. This preserves source traceability without allowing attractive vocabulary to substitute for a collision certificate.

1.2 Cradle-to-grave means lifecycle, not only algorithm

The package treats a collision engine as a lifecycle artifact:

1. establish provenance and claim boundaries;
2. define supported geometry, motion, units, tolerances, and hazards;
3. formalize the mathematics and proof obligations;
4. build a deterministic reference implementation;
5. validate against exact and adversarial cases;
6. port only verified semantics to accelerated hardware;
7. integrate detection with a separate response solver and application policy;
8. deploy with telemetry, incident replay, and explicit safe stops;
9. maintain through controlled changes and regression evidence;
10. retire unsafe or obsolete backends without silent degradation.

Every stage in the ZIP has its own numbered directory. The PDF is therefore not the only product; it is the narrative and formal index to executable and operational artifacts.

1.3 Authorship and privacy

The work is attributed to Tom Klootwijk. Personal civil identifiers and dates of birth do not strengthen a mathematical proof, do not authenticate a runtime, and are not suitable cryptographic keys. The package intentionally omits them from executable state. Release authenticity should instead use repository history, reproducible builds, conventional digital signatures, and published hashes.

Source register and claim disposition

2.1 Supplied materials

The release is grounded in four supplied sources:

- S1 – original operator-chain transcript.** A 30-page progression from parity, lower-case phi, log-polar LUT, kinematics, cone sweep, projections, distributed apex, BST/L-system branching, SDF/Klein-bottle language, and later GPU, optical, quantum, and SAR proposals (Klootwijk, 2026c).
- S2 – Formal Specification v0.1.** A 26-page formalization that separates NHDF representation, PCKF dynamics, and IHCP runtime; replaces the degenerate global zero field with cell-local constraints; distinguishes data and topology parity; bounds recursion; and defines a causal update (Klootwijk, 2026b).
- S3 – commute design transcript.** A 40-page design log that identifies the missing log-polar scale/Jacobian issue and then treats SDF plus kinematics as a claimed CCD solution, including collision-response bit operations and register-level metaphors (Klootwijk, 2026a).
- S4 – spherical-Cartesian whole-substrate extension.** A 70-page extension into measurement, sensors, actuators, chemistry, biology, medicine, photonics, Cross-Kerr logic, microvasculature, lithography, direct writing, SAR, and cosmology (Klootwijk, 2026d).

2.2 Four disposition classes

Class	Meaning	Example
Preserved	The source idea already has a valid typed interpretation.	Monotonic forward time; bounded allocation; local zero-set residual.
Reinterpreted	The source insight is useful but its original mechanism or scope was too broad.	Parity becomes a hint or diagnostic rather than a universal collision governor.
Rejected	The statement conflicts with mathematics, physics, or the evidence available.	One SDF read and one sign-bit flip completely solve general CCD and collision response.
Adapter hypothesis	The idea may define a research program only after domain laws and validation are supplied.	Optical transfer, chemical event fields, biological waveguides, medical applications.

2.3 The decisive corrections

2.3.1 From global zero to pairwise contact zero

A global $F(x) \equiv 0$ has no distance or useful gradient. CCD instead uses a family of nontrivial pairwise separation functions,

$$d_{ij}(t) = \text{sep}(A_i(t), B_j(t)), \quad \mathcal{M}_{ij} = \{t \in [0, 1] : d_{ij}(t) = 0\}. \quad (2.1)$$

Positive separation means the pair is disjoint under the backend’s convention; zero denotes contact; negative values denote overlap when a signed measure exists. This is the CCD-specific realization of distributed local zero-set semantics.

2.3.2 From parity as physics to parity as bounded information

A bitwise parity flag detects an odd count of bit inversions under a reference comparison and misses every even count. In a 2022 CCD paper, “root parity” is a precise mathematical construction for cubic roots, yet the authors explicitly note that parity cannot distinguish zero roots from the rare two-root case (B. Wang, Ferguson, Jiang, et al., 2022). This is particularly instructive: parity can be valuable in CCD, but only with a clearly defined object and a known ambiguity. Version 0.4 therefore stores parity-derived scheduling metadata but never treats it as a complete collision predicate.

2.3.3 From automatic logarithmic time dilation to coordinate-aware scheduling

Log-polar coordinates can compact scale and convert rotation/scale relationships into shifts, but they do not increase physical sampling rate for free. The Jacobian and metric must be respected, and the underlying separation must remain coordinate invariant. Version 0.4 uses log-polar bins to group or prioritize work; it does not let a coordinate transform change whether two bodies collide.

2.3.4 From a sign-bit response to a separate dynamics problem

Collision detection answers when and where contact occurs. Response must use mass, inertia, normal, friction, restitution, constraints, compliance, and material failure. A sign inversion can be a toy elastic response in one dimension, not a general physical solver. The architecture therefore ends CCD at a certificate and hands that certificate to a separate response layer.

Table 2.1: Traceability from supplied sources to the v0.4 disposition.

Source	Source idea	Disposition	v0.4 destination
S1 pp.1-6	Parity-conditioned self-referential SDF engine	Reinterpreted	Local pair separation + bounded causal state; parity is diagnostic
S1 pp.8-11	Operator precedence chain	Preserved with typing	State-threaded CCD chain and explicit address/candidate resolution
S1 pp.14-15	Parity oscillation under disturbance	Preserved as failure signal	Debounce/telemetry/safe stop; no correction claim
S1 pp.16-18	SDF zero in every LUT cell	Corrected	Family of nontrivial local constraints; in CCD, $d_{ij}(t)=0$
S1 pp.19-22	Optical/quantum sub-strate	Separated	Operator-equivalence adapter only
S1 pp.22-30	GPU/Tensor/VRAM claims	Corrected	Normal exposed operations, bounded queues, CPU equivalence first

Source	Source idea	Disposition	v0.4 destination
S2 Sec.2-5	Non-degenerate local zero sets	Adopted	Foundation for pairwise contact manifolds
S2 Sec.6-8	Forward timeline, phase, bounded branching	Adopted with limitation	Causal TOI and bounded refinement; phase/golden are heuristic
S2 Sec.12	GPU reference schedule	Extended	Broad/narrow phase, compaction, certificate emission
S3 pp.2-4	Missing log-polar Jacobian	Preserved as coordinate warning	Coordinates may schedule work; geometry remains Cartesian/invariant
S3 pp.14-17	SDF+kinematics completely solves CCD in one clock	Rejected	General CCD needs broad phase, robust narrow phase, TOI isolation, degeneracy handling, certificates
S3 pp.29-30	Compare step magnitude with SDF	Partially adopted	Conditional conservative advancement with valid distance and speed bounds
S3 pp.31-36	Autonomous/public-safety use	Restricted	Collision avoidance only, with governance; no identity or force decisions
S4 pp.12-18	Universal sensors/actuators substrate	Separated	Measurement/actuation adapter with units and calibrated transfer laws
S4 pp.22-30	Chemical, biological, macro, SAR, and cosmic maps	Separated	Independent adapters; no ontological unification
S4 pp.33-48	Medical, microvascular, biophotonic, cross-Kerr claims	Rejected as therapeutic mechanism	Simulation/research hypotheses only
S4 pp.49-70	Lithography, GDSII, 3D photonics	Separated	Fabrication adapter with PDK/DR-C/metrology gates

Research context: what robust CCD actually requires

3.1 Broad phase and narrow phase

Modern CCD systems separate a conservative broad phase from an accurate narrow phase. A false-negative broad phase is fatal because the narrow phase never sees the missed pair. Exact time-of-impact datasets have shown that implementation details can invalidate methods that appear correct on paper, and that broad and narrow phases must be evaluated together (Belgrod et al., 2023). The v0.4 scaffold therefore treats candidate capacity and unsupported backends as visible states rather than silently dropping work.

3.2 Families of methods

The roadmap is backend-plural rather than method-monolithic:

- **Closed-form primitive solvers** for spheres, planes, capsules, and translational boxes where stable roots or slabs are available.
- **Conservative advancement** for convex or implicit geometry when a trustworthy distance and motion bound exist (Mirtich, 1996; Zhang et al., 2006).
- **Interval and inclusion methods** for robust polynomial or parametric-surface queries (B. Wang, Ferguson, Schneider, et al., 2021; Chen et al., 2024).
- **Feature-based mesh CCD** using vertex-face and edge-edge tests, exact predicates, degeneracy handling, and adjacency filters.
- **Contact-integrated methods** such as IPC and its codimensional/rigid variants, which depend on collision queries but also solve a larger time-stepping problem (Li et al., 2021; Ferguson et al., 2021).
- **SDF-specific methods.** Recent work treats triangle-SDF contact directly and, in 2026, SDF-SDF time of impact through alternating spatial-temporal optimization (Pelletier-Guénette et al., 2025; Nielsen and Erleben, 2026).
- **Convex cone/ray casting and hardware acceleration.** C5D and GPU ray-tracing research illustrate active directions without implying that one representation dominates every workload (Yuan et al., 2025; Sui et al., 2024).

3.3 The reference does not compete with those methods yet

The bundled Python implementation is intentionally modest. It validates the state machine, certificate semantics, conservative-advance proof obligation, failure preservation, deterministic traces, and lifecycle tooling. It is not presented as a replacement for Tight Inclusion, IPC Toolkit, or a production physics engine. Those projects supply the correct next comparison points; IPC Toolkit itself exposes broad- and narrow-phase CCD for linear and nonlinear trajectories and explicitly states that it is not a complete physics solver (IPC-Sim Project, 2026).

Part II

CCD formal core

Problem statement and outcome semantics

4.1 Scene and motion

Let a scene contain bodies $B_i = (S_i, M_i)$, where S_i is a geometric shape and M_i maps normalized step time $t \in [0, 1]$ to a configuration. The normalized interval avoids binding the mathematics to a particular frame duration; physical time is recovered by $\tau = \tau_n + t\Delta\tau_n$.

For a pair (i, j) , a narrow-phase backend provides or derives:

1. a separation or inclusion object $d_{ij}(t)$;
2. witness features or points when meaningful;
3. a contact normal or normal set when meaningful;
4. a conservative bound on change over an interval;
5. a procedure for isolating the earliest contact.

The first time of impact is

$$t_{\text{impact}} = \inf\{t \in [0, 1] : d_{ij}(t) \leq 0\}, \quad (4.1)$$

with conventions adjusted for a minimum separation or thickness. If the pair overlaps at $t = 0$, the query is not reported as an ordinary future hit.

4.2 Seven outcomes

Table 4.1: Normative certificate outcomes.

Status	Meaning	Caller permission and obligation
HIT	Contact is found within the interval. Exact backends return a point; iterative backends return a conservative bracket.	Advance no later than the certified upper TOI, group simultaneous contacts, and invoke response.
NO_HIT	The supported backend certifies positive separation throughout the interval.	The caller may advance through the step for that pair.
INITIAL_OVERLAP	Separation is negative at the start.	Roll back, depenetrate, or invoke a dedicated overlap solver. Do not pretend the overlap began later.
INCONCLUSIVE	A possible contact could not be safely classified within numerical or resource bounds.	Reduce the step, increase precision/budget, or switch backend.

Status	Meaning	Caller permission and obligation
UNSUPPORTED	No declared backend covers the shape/motion pair.	Route to another backend or reject the scene.
INVALID_INPUT	A value, tolerance, bound, or oracle contract is invalid.	Quarantine the query and fail safe.
CAPACITY_EXCEEDED	A bounded pool or queue could not contain required work.	Preserve the required count and reconfigure; never reinterpret as no-hit.

4.3 Collision certificate

A certificate is the minimum auditable output:

$$\mathcal{C}_{ij} = (s, [t^-, t^+], \mathcal{W}, n, b, k, \mathcal{T}, h), \quad (4.2)$$

where s is status, $[t^-, t^+]$ is the TOI bracket, $\mathcal{W}$ contains witnesses, n is a normal, b names the backend, k is iteration count, $\mathcal{T}$ is bounded telemetry, and h is a deterministic digest. The digest is for replay integrity, not a replacement for a signed release or evidence chain.

Typed, state-threaded operator chain

5.1 Why the older composition needed repair

The v0.1 chain correctly ordered log-polar encoding, local boundary projection, parity, routing, kinematics, sweep, projection, and update. Its written function signatures, however, did not literally compose: an address is not a candidate cell, a bit is not a branch tree, and a tree is not a trajectory. Version 0.4 threads one enriched state through each stage.

5.2 Common state

$$\Sigma = (\mathcal{S}, I_t, \mathcal{B}, \mathcal{P}, \mathcal{O}, \mathcal{Q}, \mathcal{C}, \mathcal{H}, \mathcal{F}), \quad (5.1)$$

where $\mathcal{S}$ is the scene, I_t the active time interval, $\mathcal{B}$ swept bounds, $\mathcal{P}$ candidate pairs, $\mathcal{O}$ separation oracles, $\mathcal{Q}$ bounded refinement queues, $\mathcal{C}$ certificates, $\mathcal{H}$ telemetry/history, and $\mathcal{F}$ failures.

Every operator has type $O_k : \Sigma \rightarrow \Sigma$. The normative chain is

$$\Sigma' = U \circ \Pi_{\text{cert}} \circ R_{\text{refine}} \circ A_{\text{CA}} \circ B_{\text{bound}} \circ C_{\text{pair}} \circ E_{\text{scene}}(\Sigma) \quad (5.2)$$

with optional source-traceable scheduling metadata H_{NHDF} that may modify priority but not classification.

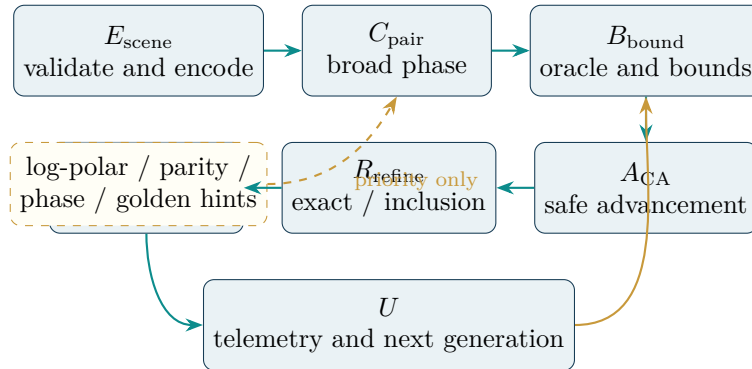


Figure 5.1: State-threaded CCD chain. Dashed NHDF hints are deliberately outside the correctness authority.

5.3 Operator obligations

Operator	Must produce	Must never do
E_{scene}	Valid units, finite values, normalized time, declared shape and motion profiles.	Silently clamp invalid values into apparently valid geometry.
C_{pair}	Conservative candidates and explicit overflow.	Prune on an unverified learned, parity, or log-polar heuristic.
B_{bound}	A backend contract: distance/inclusion object, witnesses, and valid motion/error bounds.	Call an approximate field a signed distance without a conservative error envelope.
A_{CA}	Certified no-skip time advances or an explicit stall/failure.	Force a minimum step larger than the safe bound.
R_{refine}	Earliest-contact isolation under exact, interval, or conservative semantics.	Convert budget exhaustion into no-hit.
Π_{cert}	Complete status, bracket, backend, tolerance, trace, and digest.	Hide initial overlap or unsupported geometry.
U	Deterministic state/telemetry update.	Change a geometric result after certification.

Exact primitive backends

6.1 Linear sphere–sphere

Let $p = c_B(0) - c_A(0)$, $v = v_B - v_A$, and $R = r_A + r_B$. Contact solves

$$\|p + tv\|^2 = R^2, \quad (6.1)$$

which yields $at^2 + bt + c = 0$ with

$$a = v \cdot v, \quad b = 2p \cdot v, \quad c = p \cdot p - R^2. \quad (6.2)$$

The earliest root in $[0, 1]$ is the TOI. The implementation checks initial overlap before root solving and uses a stable quadratic formulation. This exact backend is the main ground truth for the bundled randomized benchmark.

6.2 Linear sphere–plane

For a one-sided plane $n \cdot x = o$ and a sphere center $c(t)$,

$$d(t) = n \cdot c(t) - o - r. \quad (6.3)$$

Under linear relative translation, d is linear. A negative initial value is overlap; a nonnegative derivative while $d(0) > 0$ is a no-hit certificate; otherwise $t = -d(0)/\dot{d}$.

6.3 Swept axis-aligned boxes

For each axis, relative box centers and summed half-extents produce a one-dimensional overlap interval. The boxes collide if and only if the three slab intervals overlap in time. The maximum entry time is the TOI and identifies an entry normal. The backend is exact only while the boxes remain axis-aligned and motion is translational linear.

6.4 Why exact primitives matter in a general architecture

A general engine should not force every query through one iterative method. Closed-form or slab solutions are cheaper, easier to validate, and excellent differential oracles. Version 0.4 therefore dispatches by typed pair and motion profile before using conservative advancement.

Conservative advancement as the SDF bridge

7.1 The useful part of the source intuition

The commute transcript’s central CCD intuition is recognizable: if the current distance is trustworthy and the maximum rate of closure is bounded, the engine may advance by a fraction of distance divided by speed without passing through contact. This is conservative advancement, not a new one-clock law (Mirtich, 1996; Zhang et al., 2006).

7.2 No-skip theorem

Assume $d : [a, b] \rightarrow \mathbb{R}$ is continuous and L -Lipschitz,

$$|d(t) - d(s)| \leq L|t - s|, \quad (7.1)$$

where $d(t) > 0$ denotes separation. At time t_k , choose

$$h_k = \eta \frac{d(t_k)}{L_k}, \quad 0 < \eta \leq 1, \quad (7.2)$$

with L_k valid on the interval to be covered. For any $0 \leq s < h_k$,

$$d(t_k + s) \geq d(t_k) - L_k s > d(t_k)(1 - \eta) \geq 0. \quad (7.3)$$

Thus the open interval is collision-free. Repeated safe advances cannot skip a first contact, provided the distance is not overestimated and the speed bound is not underestimated.

The two contracts that cannot be waived

A sampled texture, neural implicit field, approximate Euclidean distance transform, or interpolated SDF may overestimate true separation. A velocity bound that ignores rotation, deformation, acceleration, or calibration error may underestimate closure. Either violation destroys the no-skip guarantee. In that case the engine must add a conservative bias/error bound or return `UNSUPPORTED/INCONCLUSIVE`.

7.3 SDF specialization

For a true signed distance F with spatial Lipschitz constant 1 and a moving point $x(t)$ with $\|\dot{x}\| \leq V$, the composed separation $d(t) = F(x(t)) - r$ is V -Lipschitz. If the field moves, a conservative relative-speed bound is used. If the field’s Lipschitz bound is K , then $L \leq KV_{\text{rel}}$. The executable reference supports point/sphere versus an implicit field under translation through exactly this contract.

7.4 Termination and brackets

The reference stops when distance is within a declared tolerance and returns the previous and current sample as a conservative contact bracket. If the safe step falls below temporal resolution before the distance tolerance is reached, the result is **INCONCLUSIVE**; it does not force a larger step. If the current distance minus the speed bound times remaining time stays positive, the algorithm returns a no-hit certificate for the rest of the interval.

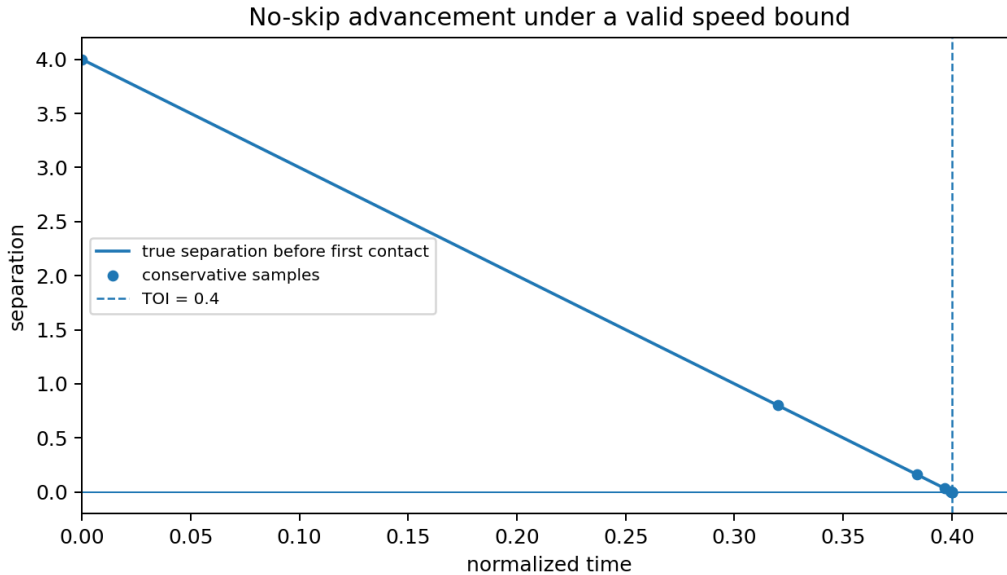


Figure 7.1: Reference conservative advancement toward a unit-sphere SDF. Sample spacing contracts near contact; no free “time dilation” is claimed.

7.5 Pseudocode

Listing 7.1: Semantics of conservative advancement in v0.4.

```

1 sample = oracle.evaluate(t)
2 if sample.distance < -distance_tolerance:
3     return INITIAL_OVERLAP
4 if sample.distance <= distance_tolerance:
5     return HIT at t
6
7 while budget remains:
8     L = oracle.closure_speed_bound(t, t_end)
9     if L <= speed_epsilon:
10        return NO_HIT
11    if sample.distance - L * (t_end - t) > 0:
12        return NO_HIT with remaining-interval certificate
13    h = safety_factor * sample.distance / L
14    if h <= time_tolerance:
15        return INCONCLUSIVE
16    previous_t = t
17    t = t + h
18    sample = oracle.evaluate(t)
19    if sample.distance <= distance_tolerance:
20        return HIT bracket [previous_t, t]
21 return INCONCLUSIVE

```

Broad phase and bounded work

8.1 Conservative swept bounds

For a body with translational motion, each coordinate is bounded over the step, including a quadratic extremum where velocity crosses zero. Shape extents expand that center interval into a swept AABB. A sweep-and-prune pass sorts by minimum x , maintains an active list, and tests overlap on all axes. Infinite planes are conservatively paired with bounded bodies.

8.2 Capacity is part of semantics

A finite machine has finite candidates, nodes, trace records, and output capacity. Let N_c be generated candidates and $N_c^{\max}$ the configured capacity. The invariant is not “overflow cannot happen”; it is

$$N_c \leq N_c^{\max} \quad \text{or the result is CAPACITY_EXCEEDED.} \quad (8.1)$$

A silent truncation would be indistinguishable from a false-negative broad phase.

8.3 Bounded interval refinement and the BST mapping

The source BST/L-system becomes a bounded earliest-first refinement tree. For interval $I = [a, b]$, sample the midpoint m and use a valid Lipschitz bound L_I :

$$\underline{d}(I) = d(m) - \frac{L_I}{2}(b - a). \quad (8.2)$$

If $\underline{d}(I) > 0$, the interval is collision-free. Otherwise it is sampled or subdivided until contact, certification, time tolerance, or node capacity. The split may be midpoint or a golden-ratio point as an experiment. The lower bound, not the aesthetic of the tree, supplies correctness.

8.4 Branching ablation rule

The golden-ratio policy **MUST** be compared with midpoint, adaptive, and backend-specific policies at equal node budget. If it does not improve a predeclared metric, it remains a source-traceable curiosity rather than a justified operator.

Log-polar, phase, parity, and topology in the CCD profile

9.1 Log-polar hint

The reference computes scheduling metadata from nonnegative separation and relative approach direction,

$$\rho = \log(1 + d), \quad \theta = \text{atan2}(v_y, v_x), \quad (9.1)$$

quantizes (ρ, θ, t) , packs a word, and records its bit parity. A priority such as $(d + \epsilon)^{-1}$ can place near contacts earlier in a work queue. The hint can improve coherence or load balance; it cannot delete a pair or certify no-hit.

9.2 Second phase difference

For a directional phase sequence ϕ_n , the source's $\Delta\Delta\phi$ becomes

$$\Delta^2\phi_n = \text{wrap}(\phi_n - 2\phi_{n-1} + \phi_{n-2}). \quad (9.2)$$

It may predict rapidly changing approach direction and request a stronger backend or smaller global step. It is not a bound unless a separate derivation relates it to motion.

9.3 Parity firewall

Normative parity firewall

Enabling or disabling NHDF parity/log-polar hints must not change the supported collision classification or certificate validity. It may change query order, iteration budget escalation, diagnostics, or performance. The bundled test suite checks classification invariance for the SDF example.

9.4 Klein-bottle topology

A Klein-bottle quotient can model non-orientable chart transitions in a specialized geometry system, but ordinary Euclidean collision detection does not require it. If a simulation explicitly uses a non-orientable manifold, chart metadata must be handled by the geometry backend. An orientation flip is not equivalent to a payload error, collision, or response impulse.

Contact output and the boundary with response

10.1 Witnesses and normals

A hit certificate may carry points w_A, w_B and a normal n . For sphere–sphere contact, n follows the center difference and the witnesses lie at radii along the normal. For implicit fields, the gradient supplies a local normal only if it is well-conditioned and consistent with the field’s sign convention. Degenerate or multiple contacts may require a normal set or contact manifold rather than one vector.

10.2 Simultaneous contacts

Let t_* be the earliest upper TOI across pairs. A response layer should group all contacts satisfying

$$t_{ij}^- \leq t_* + \tau_{\text{group}} \quad (10.1)$$

under a declared tolerance. Processing one pair and advancing past another simultaneous contact can create order-dependent penetrations.

10.3 Detection is not response

A frictionless point-mass reflection may use

$$v^+ = v^- - (1 + e)(v^- \cdot n)n, \quad (10.2)$$

for restitution e , but rigid bodies require angular momentum and effective mass; friction needs tangential constraints; resting contact and stacking need persistent constraints or potentials; deformable bodies need constitutive models. CCD supplies the event and geometry, not these laws.

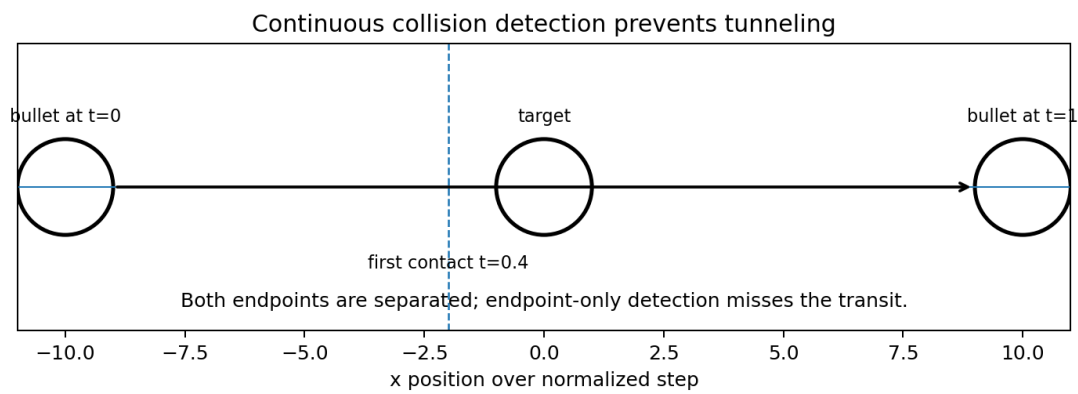


Figure 10.1: A sphere can cross another sphere while both frame endpoints remain separated. CCD isolates the first contact instead of responding to an already penetrated final state.

Part III

Executable reference and verification

Reference implementation architecture

11.1 Package structure

Module	Responsibility
<code>vector.py</code>	Minimal deterministic three-dimensional vector operations.
<code>motion.py</code>	Static, linear, and quadratic translational motion with speed and coordinate bounds.
<code>shapes.py</code>	Sphere, axis-aligned box, plane, point, and implicit-SDF types.
<code>oracles.py</code>	Separation samples, witnesses, normals, and closure-speed bounds.
<code>exact.py</code>	Exact linear primitive backends.
<code>conservative.py</code>	Contract-driven conservative advancement.
<code>interval.py</code>	Bounded midpoint/golden interval-refinement ablation.
<code>broadphase.py</code>	Swept bounds and deterministic sweep-and-prune.
<code>nhdf.py</code>	Log-polar, parity, and phase scheduling metadata outside the safety path.
<code>engine.py</code>	Typed dispatch and scene-level status reduction.
<code>types.py</code>	Configuration, status, samples, certificates, scene results, and trace hashes.
<code>io.py/cli.py</code>	Serialized scene loading and command-line execution.

11.2 Design principles embodied in code

1. **Dispatch before iteration.** Exact profiles are selected before generic advancement.
2. **Contracts are visible.** SDF Lipschitz and conservative-bias values are explicit fields.
3. **No hidden allocation.** Candidate capacity and trace capacity are bounded.
4. **No hidden fallback.** Unsupported pairs remain `UNSUPPORTED`.
5. **Deterministic replay.** A canonical JSON trace is hashed with SHA-256.
6. **Hints cannot decide.** NHDF metadata is attached after sampling and does not change the geometry path.

11.3 Exact sphere implementation excerpt

Listing 11.1: Exact linear sphere–sphere backend from the bundled Python reference.

```

1
2 def exact_sphere_sphere(body_a: Body, body_b: Body, config: CCDConfig) -> CollisionCertificate |
   None:
3     if not (isinstance(body_a.shape, Sphere) and isinstance(body_b.shape, Sphere)):
4         return None
5     va = _linear_velocity(body_a)

```

```

6     vb = _linear_velocity(body_b)
7     if va is None or vb is None:
8         return None
9     oracle = SphereSphereOracle(body_a, body_b)
10    s0 = oracle.evaluate(0.0)
11    pair = (body_a.body_id, body_b.body_id)
12    if s0.distance < -config.distance_tolerance:
13        return CollisionCertificate(CCDStatus.INITIAL_OVERLAP, pair, "exact_sphere_sphere_linear",
14                                   0.0, 0.0, s0, reason="negative initial separation").finalize()
15    if s0.distance <= config.distance_tolerance:
16        return CollisionCertificate(CCDStatus.HIT, pair, "exact_sphere_sphere_linear", 0.0, 0.0,
17                                   s0, reason="initial contact within tolerance").finalize()
18
19    p = body_b.motion.position(0.0) - body_a.motion.position(0.0)
20    v = vb - va
21    radius = body_a.shape.radius + body_b.shape.radius
22    a = v.dot(v)
23    b = 2.0 * p.dot(v)
24    c = p.dot(p) - radius * radius
25    if a <= config.speed_epsilon * config.speed_epsilon:
26        return CollisionCertificate(CCDStatus.NO_HIT, pair, "exact_sphere_sphere_linear", reason=
27                                   "zero relative speed and positive separation", sample=s0).finalize()
28    disc = b * b - 4.0 * a * c
29    if disc < 0.0:
30        return CollisionCertificate(CCDStatus.NO_HIT, pair, "exact_sphere_sphere_linear", reason=
31                                   "quadratic has no real roots", sample=s0).finalize()
32    sqrt_disc = math.sqrt(max(0.0, disc))
33    q = -0.5 * (b + math.copysign(sqrt_disc, b))
34    roots: list[float]
35    if abs(q) <= 1e-30:
36        roots = [-b / (2.0 * a)]
37    else:
38        roots = [q / a, c / q]
39    valid = sorted(t for t in roots if -config.time_tolerance <= t <= 1.0 + config.time_tolerance)
40    if not valid:
41        return CollisionCertificate(CCDStatus.NO_HIT, pair, "exact_sphere_sphere_linear", reason=
42                                   "roots lie outside the step", sample=s0).finalize()
43    toi = min(max(valid[0], 0.0), 1.0)
44    sample = oracle.evaluate(toi)
45    return CollisionCertificate(CCDStatus.HIT, pair, "exact_sphere_sphere_linear", toi, toi,
46                               sample, iterations=1, reason="earliest quadratic root").finalize()
47
48 def exact_sphere_plane(body_a: Body, body_b: Body, config: CCDConfig) -> CollisionCertificate |
49     None:
50     reverse = False
51     sphere_body, plane_body = body_a, body_b
52     if isinstance(body_a.shape, Plane) and isinstance(body_b.shape, Sphere):
53         reverse = True
54         sphere_body, plane_body = body_b, body_a
55     if not (isinstance(sphere_body.shape, Sphere) and isinstance(plane_body.shape, Plane)):
56         return None
57     vs = _linear_velocity(sphere_body)
58     vp = _linear_velocity(plane_body)
59     if vs is None or vp is None:
60         return None
61     oracle = SpherePlaneOracle(sphere_body, plane_body)
62     s0 = oracle.evaluate(0.0)
63     pair = (body_a.body_id, body_b.body_id)

```

11.4 Conservative implementation excerpt

Listing 11.2: Beginning of the conservative-advancement implementation, including explicit stall and invalid-bound behavior.

```

1
2
3 def conservative_advancement(
4     oracle: SeparationOracle,
5     config: CCDConfig,
6     *,
7     t0: float = 0.0,
8     t1: float = 1.0,
9 ) -> CollisionCertificate:
10     """Certified time advancement under an oracle-supplied Lipschitz speed bound.
11
12     Safety is conditional on two contracts:
13     1. 'evaluate(t).distance' does not overstate true separation; and
14     2. 'closure_speed_bound(a,b)' upper-bounds the absolute rate of change
15        of that separation over the interval.
16     """
17     config.validate()
18     pair = (oracle.body_a.body_id, oracle.body_b.body_id)
19     backend = "conservative_advancement"
20     trace: list[dict[str, float | int]] = []
21     t = t0
22     previous_t = t0
23     sample = oracle.evaluate(t)
24     if not math.isfinite(sample.distance):
25         return CollisionCertificate(CCDStatus.INVALID_INPUT, pair, backend, reason="non-finite
26             initial separation").finalize()
27     if sample.distance < -config.distance_tolerance:
28         return CollisionCertificate(CCDStatus.INITIAL_OVERLAP, pair, backend, t, t, sample,
29             reason="negative initial separation").finalize()
30     if sample.distance <= config.distance_tolerance:
31         return CollisionCertificate(CCDStatus.HIT, pair, backend, t, t, sample, reason="initial
32             contact within tolerance").finalize()
33
34     metadata: dict[str, object] = {}
35     for iteration in range(1, config.max_iterations + 1):
36         speed_bound = float(oracle.closure_speed_bound(t, t1))
37         if not math.isfinite(speed_bound) or speed_bound < 0.0:
38             return CollisionCertificate(CCDStatus.INVALID_INPUT, pair, backend, iterations=
39                 iteration, reason="invalid closure-speed bound", trace=trace).finalize()
40         if config.use_nhdf_hints:
41             metadata["nhdf_hint"] = nhdf_hint_for_sample(sample, oracle, t).to_dict()
42         if speed_bound <= config.speed_epsilon:
43             return CollisionCertificate(
44                 CCDStatus.NO_HIT,
45                 pair,
46                 backend,
47                 sample=sample,
48                 iterations=iteration,
49                 reason="positive separation with zero closure-speed bound",
50                 trace=trace,
51                 metadata=metadata,
52             ).finalize()
53
54     remaining = t1 - t
55     safe_step = config.safety_factor * sample.distance / speed_bound
56     no_hit_margin = sample.distance - speed_bound * remaining
57     trace_item = {
58         "iteration": iteration,

```

```

55         "t": t,
56         "distance": sample.distance,
57         "speed_bound": speed_bound,
58         "safe_step": safe_step,
59         "remaining": remaining,
60         "no_hit_margin": no_hit_margin,
61     }
62     if len(trace) < config.max_trace_steps:
63         trace.append(trace_item)
64
65     if no_hit_margin > 0.0:
66         return CollisionCertificate(
67             CCDStatus.NO_HIT,
68             pair,
69             backend,
70             sample=sample,
71             iterations=iteration,
72             reason="Lipschitz no-hit certificate covers the remaining interval",
73             trace=trace,
74             metadata=metadata,
75         ).finalize()
76
77     if safe_step <= config.time_tolerance:
78         # The query is closer than the declared temporal resolution. Returning
79         # INCONCLUSIVE is safer than silently stepping past a possible contact.
80         return CollisionCertificate(
81             CCDStatus.INCONCLUSIVE,
82             pair,
83             backend,
84             toi_lower=max(previous_t, t0),
85             toi_upper=min(t + config.time_tolerance, t1),
86             sample=sample,
87             iterations=iteration,
88             reason="advancement stalled at temporal resolution before contact tolerance",

```

11.5 C++ and GPU slices

The C++17 directory implements the exact sphere–sphere query and builds without external dependencies. The CUDA directory contains a source-only batched exact sphere kernel. It was not compiled in the release environment because no CUDA compiler or target GPU was available. The report therefore makes no CUDA performance or equivalence claim.

Test plan and included results

12.1 Test layers

The package contains 22 Python tests spanning exact roots, high-speed tunneling, no-hit, initial overlap, AABB slabs, true-SDF conservative advancement, sphere radius against an SDF, quadratic translation, invalid Lipschitz values, hint invariance, interval split policies, broad-phase pruning, capacity failure, unsupported pairs, deterministic digests, and invalid configuration. A C++ smoke test independently checks the sphere–sphere TOI.

12.2 Generated differential benchmark

The reproducible benchmark generates 5,000 linear sphere–sphere queries with a fixed seed. The exact quadratic backend is treated as ground truth and compared with the conservative-advancement oracle. It also evaluates a deliberately weak detector that checks only $t = 0$ and $t = 1$.

Table 12.1: Bundled generated benchmark summary. Timings are machine-specific CPython measurements and not product claims.

Metric	Result
Queries	5,000
Exact hits	2,502
Hits missed by endpoint-only discrete check	2,500
Conservative false negatives	0
Conservative false positives	0
Conservative inconclusive	0
Mean / maximum CA iterations	7.332 / 43
Mean TOI upper error	3.362e-10
Maximum TOI upper error	1.789e-08
Exact / CA queries per second	64,173 / 13,796

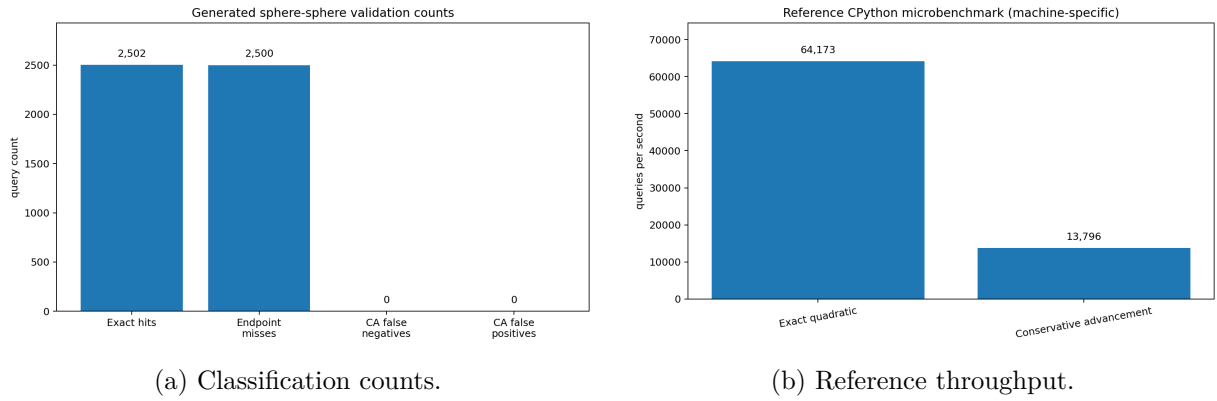


Figure 12.1: The benchmark verifies the included sphere profile and demonstrates the endpoint-tunneling failure. It does not validate mesh or GPU completeness.

12.3 Interpretation

The zero false-negative result is evidence that the implementation matches its exact oracle on this generated distribution. It is not a statistical proof over all inputs. The next gate must use exact corpora containing millions of realistic and near-degenerate feature queries, as advocated by the CCD benchmark literature (B. Wang, Ferguson, Schneider, et al., 2021; Belgrod et al., 2023).

Robustness and adversarial validation plan

13.1 Mandatory geometry cases

1. tangential contact and repeated roots;
2. near-parallel edge–edge and near-coplanar vertex–face motion;
3. contact at exactly $t = 0$ or $t = 1$;
4. zero-area triangles, duplicate vertices, and zero-length edges;
5. coincident features and ambiguous normals;
6. scale extremes and mixed units;
7. rapidly rotating shapes whose closest feature changes;
8. accelerated and nonlinear trajectories;
9. adjacency and self-collision filters;
10. multiple simultaneous impacts.

13.2 Mandatory software and resource cases

1. NaN, infinity, signed zero, overflow, underflow, and denormal behavior;
2. compiler and optimization differences;
3. deterministic tie ordering;
4. candidate, queue, trace, and output exhaustion;
5. stale generation reads and ring-buffer wrap;
6. CPU/GPU disagreement;
7. corrupted serialized scenes and invalid schemas;
8. approximate SDFs with known positive and negative error.

13.3 Differential program

Version 0.5 should ingest public exact TOI data and compare at least three independent paths: an exact/inclusion reference, the NHDF-CCD backend under test, and a second established library. It should report false negatives, false positives, TOI interval width, runtime distribution, memory, and failure status by case family. Test results should remain linked to release and compiler hashes.

Falsifiable NHDF hypotheses inside CCD

14.1 Core hypotheses

Table 14.1: CCD-centered falsifiable hypotheses.

ID	Hypothesis	Required comparison
H-CCD-1	Log-polar bins reduce divergence or improve cache/work-queue locality without changing classification.	Equal queries, equal backend, linear/Cartesian or distance-only bucketing.
H-CCD-2	A parity-derived anomaly signal predicts queries that require escalation.	Threshold, residual magnitude, random bit, and no-hint baselines; ROC and cost.
H-CCD-3	$\Delta^2\phi$ improves motion-bound selection for rapidly changing approach direction.	Actual orientation/curvature bounds and fixed conservative bounds.
H-CCD-4	Golden-ratio interval splitting reduces nodes or time for a declared workload.	Midpoint, secant, adaptive, and earliest-first strategies at equal correctness.
H-CCD-5	Bounded self-referential feedback improves next-frame broad-phase coherence.	Open-loop refit/rebuild and standard temporal coherence baselines.
H-CCD-6	SDF-specific backends improve quality-per-cost for complex implicit geometry.	Triangle mesh, convex proxy, and recent triangle-SDF/SDF-SDF methods.
H-CCD-7	GPU implementation preserves all CPU statuses and TOI bounds while improving throughput or latency.	Deterministic replay on exact and adversarial corpora.

14.2 Kill criteria

An NHDF-specific operator should be removed from the CCD path when it fails equal-budget ablations, increases false negatives, widens TOI bounds without compensating value, creates excessive divergence, or cannot be explained independently of the source metaphor. The architecture is stronger when unnecessary operators are deleted.

Part IV

Hardware, integration, and lifecycle

GPU roadmap

15.1 Semantics first

GPU acceleration begins only after a CPU backend passes exact differential tests. The GPU is not asked to execute “topology” or “SDF zero” by name; it executes concrete bounds, roots, interval predicates, scans, queues, distance queries, and reductions. Matrix or ray hardware is used only where exposed semantics match the subproblem.

15.2 Proposed pipeline

1. upload or generate structure-of-arrays body state;
2. compute swept bounds;
3. run a conservative broad phase;
4. compact candidates with prefix sums while returning required count and overflow;
5. bucket by backend and motion type;
6. execute batched exact primitive kernels;
7. execute robust mesh/inclusion work queues;
8. reduce earliest TOI with a deterministic tie policy;
9. emit certificates and bounded traces;
10. replay selected samples on CPU.

15.3 Floating-point policy

Fast-math modes can alter predicate signs and root intervals. The GPU profile must classify operations into performance-tolerant arithmetic and correctness-critical predicates. A faster kernel that silently changes HIT to NO_HIT is not an optimization. Exact/inclusion methods and outward rounding may be required for robust backends.

15.4 NHDF scheduling experiments

Log-polar separation/approach bins could group similar iteration counts; $\Delta^2\phi$ could identify changing motion; parity could request replay; golden splits could shape work queues. Each experiment must preserve the candidate set and certificate semantics. The source CUDA/Tensor claims are therefore replaced by measurable scheduling hypotheses rather than assertions that Tensor Cores implement parity or topology.

15.5 Acceptance metrics

- zero false negatives on declared exact corpora;
- CPU/GPU status equality;
- TOI bracket containment and width;
- overflow and unsupported-state equality;
- latency percentiles, not only average throughput;
- memory and queue high-water marks;
- energy and thermal behavior on the actual device;
- deterministic mode and documented non-deterministic high-throughput mode.

Application integration and response boundary

16.1 Step protocol

A conservative event-driven simulation step is:

1. predict but do not commit motion;
2. find candidates and query CCD;
3. select the earliest conservative TOI;
4. group near-simultaneous contacts;
5. advance to the allowed time;
6. solve response or constraints;
7. repeat for the remaining substep under a bounded event count;
8. commit state and telemetry.

16.2 Rollback and safe stop

The caller must have policies for initial overlap, inconclusive queries, unsupported shapes, and capacity exhaustion. In safety-relevant systems, the default should be reduced speed, reduced step, larger clearance, or stop – not an optimistic no-hit.

16.3 Public-safety and autonomous-agent boundary

CCD has legitimate uses in emergency vehicles, robots, drones, camera rigs, and mechanical safety. It does not establish intent, identity, guilt, threat, or proportionate force. A policing adapter must include lawful purpose, data minimization, human control over consequential decisions, independent privacy/bias/security review, conventional evidence cryptography, uncertainty display, and safe-stop behavior.

Cradle-to-grave gates

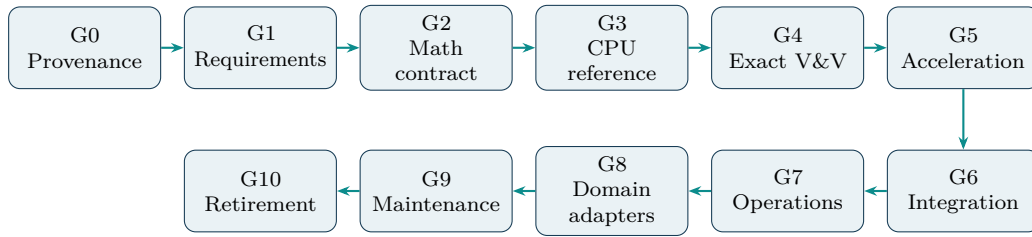


Figure 17.1: Lifecycle gates. Flow runs left-to-right on the first row and right-to-left on the second. An adapter may enter only after the core evidence it depends on exists; retirement remains a designed stage.

17.1 Gate G0 – provenance and claim boundary

Input: supplied documents and external research. **Output:** source register, traceability matrix, privacy policy, and disposition classes. **Pass:** every statement is sourced, reasoned, or labeled hypothesis.

17.2 Gate G1 – requirements and hazards

Declare shapes, motion class, clearance, units, tolerances, missed-contact cost, false-positive cost, determinism, latency, and caller failure policy. **Pass:** supported and unsupported profiles are unambiguous.

17.3 Gate G2 – mathematical contract

Define separation/inclusion objects and valid error/motion bounds. **Pass:** every no-hit statement has a proof obligation and heuristics have no hidden safety authority.

17.4 Gate G3 – deterministic CPU reference

Implement the smallest clear semantics. **Pass:** unit/property tests, reference vectors, schemas, failure preservation, and deterministic replay.

17.5 Gate G4 – robust differential validation

Use exact/adversarial data and independent implementations. **Pass:** zero false negatives in the declared profile and bounded TOI error. This release has not passed G4 for triangle meshes.

17.6 Gate G5 – accelerated backend

Port verified semantics to C++/GPU/other hardware. **Pass:** equivalence plus measured performance, memory, energy, and failure behavior.

17.7 Gate G6 – application integration

Add response, rollback, uncertainty, and hardware-in-loop testing. **Pass:** end-to-end tunneling and failure recovery succeed under realistic timing.

17.8 Gate G7 – deployment and operations

Add signed releases, configuration control, telemetry, alarms, incident replay, and safe disable. **Pass:** divergence, overflow, invalid bounds, and unsupported inputs are detected operationally.

17.9 Gate G8 – domain adapters

Only now map the core into a domain with its own laws. **Pass:** units, invariants, uncertainty, and baselines are preserved, and operator equivalence is measured.

17.10 Gate G9 – maintenance

Every predicate, tolerance, compiler mode, queue, and broad-phase change triggers regression evidence. **Pass:** change record, tests, migration, and rollback exist.

17.11 Gate G10 – retirement

Freeze unsafe or obsolete backends, migrate callers, archive reproducibility artifacts, revoke deployment, and delete personal/sensor data according to policy. **Pass:** no silent fallback to discrete endpoint checks.

Operations, incident response, and change control

18.1 Runtime telemetry contract

At minimum, record

$$\{g, \Delta t, N_b, N_c, N_{\text{backend}}, [t^-, t^+], d_{\text{min}}, k, \tau_d, \tau_t, N_{\text{overflow}}, N_{\text{unsupported}}, N_{\text{inconclusive}}, T_{\text{run}}, h\}. \quad (18.1)$$

This separates geometry, numerical, resource, and application failures.

18.2 Alarm conditions

- any replayed false negative;
- repeated inconclusive results at an unacceptable rate;
- capacity high-water or overflow;
- NaN/Inf or invalid motion/error bound;
- CPU/GPU divergence;
- unsupported shape/motion entering a safety path;
- trace or configuration mismatch.

18.3 Incident package

Preserve the exact scene, units, configuration, build and compiler hashes, hardware/runtime versions, certificate, bounded trace, random seed, and caller action. Screenshots or identity metadata are not substitutes for cryptographic and reproducible evidence.

18.4 Change categories

A documentation typo is minor. A change to a tolerance, broad-phase bound, root formula, field interpolation, floating-point mode, queue capacity, tie order, or motion model is correctness-affecting and requires revalidation. A new adapter is not “just configuration” if it changes state, units, or invariants.

Table 18.1: Risk register summary.

ID	Risk	Severity	Likelihood	Primary mitigation
R-001	False-negative broad phase	Critical	Medium	Use conservative swept bounds; exact dataset; never use learned/parity pruning without verification
R-002	Distance oracle overestimates true separation	Critical	Medium	Require true/conservatively biased distance and calibrated Lipschitz/error bounds
R-003	Closure-speed bound too low	Critical	Medium	Derive per motion/shape; test acceleration/rotation; fail invalid bounds
R-004	Floating-point degeneracy	High	High	Inclusion/exact predicates; scale normalization; adversarial corpus; multiple precision modes
R-005	Iteration or queue exhaustion	High	Medium	Explicit INCONCLUSIVE/CAPACITY_EXCEEDED; step reduction or robust backend
R-006	Contact response conflated with detection	High	Medium	Separate APIs and certificates; response solver owns impulses/friction
R-007	Parity/golden topology treated as proof	High	High	Keep outside safety authority; mandatory ablations
R-008	GPU fast-math changes signs	Critical	Medium	Predicate-specific compile modes; CPU replay; disable backend on divergence
R-009	Approximate/neural SDF lacks error envelope	Critical	High	Do not certify; add conservative error bound or mesh/exact fallback
R-010	Medical or public-safety overclaim	Critical	Medium	Simulation-only labels, governance, clinical/regulatory evidence gates
R-011	Personal identifiers used as keys	High	Medium	Omit from runtime; use standard cryptography and release hashes
R-012	Domain adapter erases constitutive laws	High	High	Typed adapter contract with units/invariants/baselines

Part V

Separated domain adapters

Uniform adapter contract

19.1 Definition

A domain adapter is

$$\mathcal{A}_d = (X_d, \Phi_d, g_d, E_d, O_d, D_d, I_d, M_d, F_d), \quad (19.1)$$

where X_d is domain state, Φ_d accepted evolution, $g_d = 0$ an event/contact/constraint surface, E_d, D_d encoder/decoder, O_d the implemented operator subset, I_d units and invariants, M_d metrics/baselines, and F_d failures/stops.

A meaningful equivalence claim requires

$$\sup_{x \in \mathcal{D}_d} \|D_d(\mathcal{N}_{\Delta t}(E_d(x))) - \Phi_d^{\Delta t}(x)\|_{W_d} \leq \epsilon_d \quad (19.2)$$

with calibrated uncertainty and declared domain $\mathcal{D}_d$. Reusing the symbols ρ , ϕ , parity, SDF, or tree does not prove this inequality.

19.2 Relationship classes

- **Literal CCD:** moving geometric bodies and contact.
- **Boundary-event reuse:** a continuous threshold crossing uses bracketing/inclusion logic but is not called geometric collision.
- **Scheduling analogy:** queues, coordinates, and feedback are reused without physical equivalence.
- **Research hypothesis:** no transfer equivalence has yet been demonstrated.

Table 19.1: Separated domain adapters and evidence gates.

Adapter	CCD relationship	Maturity	Next evidence gate	Hard boundary
Graphics and game engines	Literal CCD	D2 executable scaffold	Exact/inclusion CCD datasets; response integration; GPU equivalence	Parity/golden/Klein are optional scheduling meta-data
Robotics and public safety	Literal CCD	D1 architecture map	Certified motion bounds, sensor uncertainty, safe stop, governance	No automated identity, suspicion, or use-of-force authority

Adapter	CCD relationship	Maturity	Next evidence gate	Hard boundary
Measurement, sensors, actuators	Literal contact plus boundary-event reuse	D1 contract	Traceability, uncertainty budget, actuator dynamics, hardware-in-loop tests	SDF is a calibrated geometric model, not the material itself
SAR and wave sensing	Boundary-event reuse	D0-D1 hypothesis	Measured focus metrics against established autofocus baselines	A focus zero set is not a rigid-body collision
GPU and edge AI	Scheduling analogy plus literal CCD acceleration	D1 design	Exact replay, no-false-negative broad phase, quality/performance baseline	Learned or parity proposals may not prune safety candidates
Optical and photonic computing	Operator-equivalence research	D0-D1 hypothesis	Calibrated transfer error, drift, bandwidth, energy, repeatability	Propagation direction is causal order, not zero latency
Quantum and cross-Kerr	Research hypothesis	D0	Gate/process tomography, noise model, fidelity, classical baseline	Ideal cross-Kerr parity is not assumed in biology or hardware
Chemical reaction networks	Boundary-event reuse	D0-D1 hypothesis	Mass/charge balance, calibrated rate laws, reactor experiments	Equilibrium is not automatically an SDF and parity is not universal
Biology and medicine	Simulation-only boundary events	D0	Preclinical/clinical evidence and regulatory pathway for any intervention	No treatment, regeneration, dementia reversal, or crash-immunity claim
Fabrication and 3D photonics	Literal geometry checks plus design rules	D1 map	PDK/DRC compliance, optical simulation, metrology, fabricated test coupons	Polar coordinates do not bypass fabrication tolerances
Cosmology and holography	Analogy only	D0	Derive a real duality and compare predictions	SDF boundaries and projection do not establish AdS/CFT-like holography

Graphics, robotics, measurement, and public safety

20.1 Graphics and game engines

This is the direct application. The pairwise contact zero set, swept bounds, TOI, witnesses, and response handoff are literal. Version 0.4 supports only a primitive slice, so the next work is robust triangle feature CCD, rotation, deformables, self-collision, simultaneous contacts, and engine response. Recent exact and inclusion datasets provide the required validation foundation (B. Wang, Ferguson, Schneider, et al., 2021; Belgrod et al., 2023).

The source projections – pyramid side, circle, sphere – become calibration geometry, not guaranteed outputs. Analytic cones may serve as swept support or collision cones, but they must be conservative for the actual shape and motion.

20.2 Robotics

Robot and vehicle planning uses CCD to validate link and environment trajectories. Motion uncertainty, control latency, localization error, and braking distance must enlarge the swept geometry or enter the bound. Hardware-accelerated ray-tracing research is promising for batched robotic collision queries, but representation accuracy and conservative coverage remain explicit tradeoffs (Sui et al., 2024).

20.3 Measurement, sensors, and actuators

The whole-substrate document’s broad measurement chain is retained in disciplined form: measurand, transduction, indication, calibration, uncertainty, and interpretation. Metrology terminology and traceability are standardized independently of NHDF (Joint Committee for Guides in Metrology, 2012). NHDF may provide a common data/control record, while each sensor retains its transfer function, units, bandwidth, noise, drift, hysteresis, and environment.

CCD is literal for moving stages, shutters, rotors, valves, robotic arms, scanner heads, and mechanical envelopes. For temperature, pressure, or voltage limits, the interval machinery may detect a threshold crossing but should be named boundary-event detection. Actuator response follows electromechanical or material dynamics; parity may request a safety state but is not a restoring force in matter.

20.4 Public safety

A collision engine can support emergency-vehicle navigation, robot safety, mechanical camera rigs, and evidence-device integrity checks. The adapter must not extend geometric prediction into claims of hidden

intent or automatic enforcement. Conventional cryptography and chain-of-custody practice, not personal metadata or a geometric phase lock, must secure evidence.

SAR, GPU/AI, and wave sensing

21.1 SAR and ionospheric disturbance

The source maps echo phase and timing residuals into a log-polar LUT, platform motion into kinematics, and focused output into projection. Version 0.4 preserves this as a wave-sensing adapter:

$$X_{\text{SAR}} = (s_{\text{echo}}, x_{\text{nav}}, \hat{\phi}_{\text{error}}), \quad (21.1)$$

$$g_{\text{SAR}}(t) = \text{declared focus or phase-consistency residual}. \quad (21.2)$$

A conservative threshold-crossing solver may bracket when the residual leaves an admissible region; bounded trees may partition range, azimuth, scale, or hypotheses. This is not rigid-body collision and does not remove the need for propagation, calibration, autofocus, and image-quality models.

Metrics must include phase RMSE, point-target resolution, peak and integrated sidelobe ratios, image entropy/contrast, compute, memory, dropped frames, and controlled disturbance. “G3 immunity” is rejected; the correct behavior is measured degradation, safe mode, stronger integrity codes, and explicit failure.

21.2 GPU CCD

The GPU adapter is a direct acceleration map discussed in Chapter 13. It must preserve every candidate and status. Current CCD work demonstrates both scalable parallel algorithms and the importance of exact validation; the main lesson is that speed and correctness must be measured together (Belgrod et al., 2023).

21.3 Edge AI

The AI material becomes a separate codec/routing hypothesis. Log-polar coding, local constraints, mixed precision, sparse routing, and bounded feedback may be tested against established quantization, sparsity, and distillation baselines. A learned broad phase may prioritize candidates, but a safety profile cannot allow it to drop pairs unless a conservative verification layer proves the drop safe.

Optical, photonic, quantum, and Cross-Kerr adapters

22.1 Optical operator equivalence

Programmable optics can implement selected linear functions and self-configuring transfer networks (Miller, 2013). The NHDF question is therefore concrete: does a physical transfer T_{phys} reproduce a declared operator O within error,

$$\|T_{\text{phys}}(E_{\text{in}}) - O(E_{\text{in}})\| \leq \epsilon? \quad (22.1)$$

A first experiment should implement only log-polar encoding plus a local projection or constraint stage. It must report phase/amplitude error, drift, calibration, bandwidth, repeatability, and total energy including conversion.

For CCD, a photonic stage might accelerate a distance transform, convolution, or candidate ranking. It cannot certify no-hit until its worst-case error is conservatively bounded under fabrication variation, temperature, source/detector noise, aging, and calibration failure. A digital verifier should remain in the loop.

22.2 Photonic waveguides and direct writing

Curved and three-dimensional waveguides can be fabricated, but “native polar geometry” does not abolish discretization or process windows. A tool path still has finite resolution; optical performance still depends on refractive index, mode confinement, bend loss, roughness, coupling, material absorption, and packaging. The fabrication adapter in Chapter 22 defines the required sequence.

22.3 Quantum profile

A quantum implementation requires state preparation, Hilbert space, Hamiltonian/channel, gates, noise, measurement, readout, and resource scaling. A classical zero-set is not automatically a quantum code. A quantum parity measurement is not the same object as payload XOR parity, topological orientation parity, or CCD root parity.

22.4 Cross-Kerr boundary

The idealized interaction

$$H = \hbar\kappa\hat{n}_a\hat{n}_b \quad (22.2)$$

can produce a conditional phase under ideal assumptions. Traveling-wave single-photon gates face response and phase-noise limits; detailed analyses show that causality-induced noise can preclude high-fidelity π

phase shifts in important regimes (Dove et al., [2014](#)). Version 0.4 therefore treats Cross-Kerr parity as a laboratory hypothesis, not as a CCD primitive, a guaranteed chip component, or a biological mechanism.

Chemical, biological, and medical adapters

23.1 Chemical reaction networks

A chemical state is not a geometry word. It is governed by stoichiometry and kinetics, commonly

$$\dot{c} = Sv(c, t), \quad (23.1)$$

with diffusion, temperature, activities, and thermodynamics as required. A concentration threshold or phase boundary $g(c, t) = 0$ may use conservative event detection. Log scaling may be useful over wide concentration ranges. A branch graph may represent reactions or compartments.

Programmable BZ-reaction processors demonstrate that hybrid digital-chemical computation is physically real, but they use explicit reactor dynamics, cells, hysteresis, and digital control rather than a universal parity law (Sharma et al., 2024). Any NHDF chemical adapter must conserve mass/charge, preserve positivity, calibrate rate laws, and compare with standard solvers and experiments.

23.2 Biological modeling

CCD has direct roles in tissue-tool contact, joint surfaces, catheter navigation, surgical simulation, cells/devices, and anatomical mesh self-collision. Detection must be coupled to biomechanics, fluid dynamics, constitutive behavior, and uncertainty.

Vascular branching is not established as universal golden-ratio scaling. Murray's minimum-work work provides a historically important, assumption-dependent alternative model (Murray, 1926). Any proposed branching law must be dimensionally valid and tested against data.

23.3 Biophotonics

Ultraweak photon emission is measurable, and theoretical models have proposed myelinated axons as possible optical waveguides (Kumar et al., 2016). Disease-model studies have reported altered ultraweak photon activity and spectral shifts (Z. Wang et al., 2023; Sefati et al., 2024). These observations do not demonstrate neural NHDF computation, tubulin photonic-bandgap logic, mitochondrial Cross-Kerr parity, or instant rerouting around damaged tissue.

23.4 Medical stop line

No therapeutic claim

This package does not claim to prevent or reverse dementia, regenerate neural tissue, restore swallowing, bypass vascular lesions, stop sepsis, or make an organism immune to failure. A mathematical reparameterization is not an intervention. Any medical claim requires a defined device or treatment, biological mechanism, dose/exposure, toxicology, preclinical replication, clinical trials, ethics, and regulatory review. Current authoritative health information does not support a general dementia cure claim (National Institute on Aging, [2026](#)).

Fabrication and 3D photonics

24.1 What CCD contributes literally

CCD can validate moving tool heads, stages, wafer handling, robot paths, and additive-manufacturing trajectories. Static intersection and minimum-separation checks can validate waveguide crossings, ring resonators, electrodes, vias, packages, and curved mask geometry. These are concrete design-rule and motion-planning uses.

24.2 What coordinate choice does not remove

Polar or spherical parameterization may describe rings and spirals efficiently, but fabrication equipment consumes finite-resolution layouts or paths. Process design kits, minimum width/space, etch bias, line-edge roughness, overlay, stitching, material deposition, coupling, thermal drift, and metrology remain. ASML’s current High-NA EUV platform is a lithography scanner with 0.55 numerical aperture and an advertised 8 nm resolution; it is not a custom chip designer or complete fab (ASML, [2026](#)).

24.3 Credible cradle-to-device sequence

1. define optical/electrical function and process platform;
2. select materials and a foundry/process design kit;
3. perform electromagnetic and thermal simulation;
4. generate curvilinear layout with explicit approximation tolerance;
5. run design-rule and connectivity checks;
6. run lithography/process-window simulation where needed;
7. fabricate test coupons;
8. measure geometry, loss, phase, bandwidth, and drift;
9. calibrate the model and repeat;
10. only then integrate a computing or sensing demonstration.

Direct laser writing can alter the manufacturing route, but it does not “bypass fabrication” in the sense of removing materials, equipment, process calibration, packaging, or metrology. It is a different fabrication stack.

Cosmology and the limit of the substrate claim

25.1 Holographic word versus holographic principle

Projection, wavefront processing, or distributed storage may be called holographic in engineering. The holographic principle in quantum gravity is a precise relation between theories, observables, degrees of freedom, entropy, and dynamics; AdS/CFT is the canonical example (Maldacena, [1998](#)). An SDF boundary and a projection operator do not by themselves establish such a duality.

25.2 Permitted use

CCD and event-surface machinery may appear inside numerical relativity, particle simulation, ray propagation, or astrophysical software. That computational reuse does not imply that spacetime fundamentally executes the NHDF chain. The cosmology adapter remains D0 until it states two theories and derives a testable correspondence or new prediction.

Integrated adapter roadmap

26.1 Order of work

The separate materials should not all advance in parallel at equal priority. A rational sequence is:

1. complete triangle/convex/SDF CCD validation;
2. build a GPU-equivalent backend;
3. integrate one graphics or robotics application;
4. test NHDF scheduling ablations;
5. implement a narrow optical operator-equivalence experiment;
6. pursue fabrication only after simulation and process rules are selected;
7. keep chemical, biological, medical, quantum, and cosmological work as independent collaborations with domain specialists and separate evidence plans.

26.2 Why this preserves the entire project

Separating adapters is not a retreat. It gives every domain a common systems contract while preserving the constitutive laws that make the domain real. The shared contribution can be an architecture for constrained state, continuous event detection, bounded routing, observation, and feedback. A universal ontological claim is unnecessary and currently unsupported.

Part VI

Release roadmap, risks, and appendices

Version roadmap to 1.0

Table 27.1: Proposed release sequence.

Version	Center	Exit evidence
0.4	Typed CCD scaffold	Current PDF/ZIP; primitive exact/CA reference; tests; lifecycle and adapters.
0.5	Robust linear triangle CCD	Exact TOI corpus, vertex-face and edge-edge backends, degeneracy catalog, zero false negatives in declared profile.
0.6	Convex and rigid motion	GJK/support mapping or cone casting, rotational bounds, curved trajectory handling, independent differential tests.
0.7	SDF production profile	Certified/biased fields, triangle-SDF and SDF-SDF comparison, rotation, adaptive sampling, error envelopes.
0.8	GPU backend	CPU/GPU equivalence, deterministic mode, overflow equality, measured latency/memory/power.
0.9	Application proof	One graphics/robotics integration outperforming or complementing a selected baseline without correctness regression.
1.0	Stable core	Frozen API, supported profiles, external review, continuous exact-data testing, operations and retirement support.
1.x-adapter	Domain-specific	Separate version, domain laws, uncertainty, ethics/regulation, and measured operator equivalence.

27.1 Definition of done for 1.0

- public supported-profile matrix;
- exact and adversarial continuous test service;
- independent reproduction on at least two platforms;
- complete failure-state and resource semantics;
- CPU and accelerated backends with equivalence reports;
- stable serialization and API;
- one validated application integration;
- security, privacy, operations, incident, change, and retirement documentation;
- removal or isolation of operators that fail ablations.

Conclusions

The earlier work can be made stronger by narrowing its first proof target. CCD provides a literal and demanding realization of the central idea: valid states lie on local constraint surfaces; motion is continuous and causal; swept bounds conservatively restrict work; bounded refinement isolates events; projection emits an observable certificate; and feedback updates the next generation.

The executable v0.4 release demonstrates this architecture for exact linear spheres, sphere–plane contact, swept AABs, and conditional point/sphere–SDF conservative advancement. It also demonstrates the discipline that a complete engine needs: explicit unsupported cases, initial overlap, inconclusive results, capacity failure, deterministic traces, schemas, tests, benchmarks, and a lifecycle beyond the algorithm.

The source-specific log-polar, parity, phase, golden-ratio, and topology operators remain in the research program, but behind a correctness firewall. They must earn a place through ablation and cannot override geometry. This is the key turn from a broad substrate narrative to a testable computational contribution.

The optical, quantum, chemical, biological, medical, SAR, fabrication, and cosmological material has not been discarded. It has been mapped into independent adapters that preserve domain laws and define what evidence would be required. That separation is the cradle-to-grave foundation for responsible expansion: one verified core, many explicitly bounded adapters, and no silent leap from shared notation to universal physics.

Normative minimum specification

An implementation may call itself an NHDF-CCD conforming engine only if:

1. it defines continuous step time and monotonic generation semantics;
2. it defines supported shape and motion profiles;
3. it defines conservative broad-phase bounds and visible capacity behavior;
4. every supported pair has a declared exact, conservative, or inclusion backend;
5. every no-hit result is justified by a valid bound or exact predicate;
6. it distinguishes hit, no-hit, initial overlap, inconclusive, unsupported, invalid input, and capacity exceeded;
7. it outputs a TOI point or bracket with tolerance and backend identity;
8. it separates detection from response;
9. it bounds iterations, queues, traces, and workspace;
10. it supports deterministic replay for validation;
11. it does not let log-polar, parity, golden-ratio, phase, or topology hints suppress candidates or certify outcomes without a separate proof;
12. it exposes telemetry sufficient to reproduce a failure;
13. any accelerated backend is compared with a verified semantic reference;
14. any domain adapter declares state, laws, units, invariants, uncertainty, metrics, and failure conditions.

Symbol table

Table B.1: Principal symbols.

Symbol	Meaning
$B_i = (S_i, M_i)$	Body with shape and motion.
$t \in [0, 1]$	Normalized time over one proposed simulation step.
$d_{ij}(t)$	Pairwise separation or backend-specific contact function.
t_{impact}	Earliest time of impact.
$[t^-, t^+]$	Conservative time-of-impact bracket.
L	Valid Lipschitz/closure-speed bound for separation over an interval.
η	Conservative advancement safety factor in $(0, 1]$.
τ_d, τ_t	Distance and time tolerances.
Σ	Complete typed runtime state threaded through operators.
E_{scene}	Input validation and encoding.
C_{pair}	Conservative broad-phase candidate generation.
B_{bound}	Narrow-phase oracle and bound construction.
A_{CA}	Conservative advancement.
R_{refine}	Exact/inclusion/interval earliest-contact refinement.
Π_{cert}	Certificate emission.
U	Telemetry, cache, and next-generation update.
ρ, θ	Log-separation and approach-direction scheduling coordinates.
$\Delta^2 \phi$	Wrapped second finite difference of phase/direction metadata.

Package manifest overview

Table C.1: Top-level ZIP layout.

Path	Purpose
00_provenance/	Source register, evidence boundary, privacy/authorship policy, and mounted copies of sources S1–S3; source S4 remains a registered conversation attachment.
01_problem_definition/	Requirements and release acceptance criteria.
02_formal_specification/	Typed operator contract, certificate semantics, JSON schemas, and state machine.
03_reference_implementation/	Python package, examples, C++17 slice.
04_verification/	Tests, adversarial catalog, plan, and bundled verification report.
05_benchmarks/	Deterministic generators, reference vectors, JSON/CSV results.
06_gpu_port_plan/	GPU architecture and uncompiled CUDA exact-sphere skeleton.
07_integration/	Engine integration, response boundary, public-safety governance.
08_operations/	Runbook and change control.
09_retirement/	Decommissioning and migration plan.
10_domain_adapters/	Uniform contract, matrix, and eleven separated adapter documents.
11_report/	LaTeX source, figures, references, generated tables, and final PDF.
scripts/	Full verification, benchmark-asset generation, report build, and package hash checks.
SHA256SUMS.txt	Hashes for deliverable files after package verification.

Reproduction commands

Listing D.1: Full local verification.

```
1 python -m pip install -e .
2 pytest
3 python 05_benchmarks/generate_reference_vectors.py
4 python 05_benchmarks/run_benchmarks.py
5 python 11_report/generate_report_assets.py
6 cmake -S 03_reference_implementation/cpp -B build/cpp \
7     -DCMAKE_BUILD_TYPE=Release
8 cmake --build build/cpp
9 ctest --test-dir build/cpp --output-on-failure
10 bash scripts/build_report.sh
11 python scripts/package_check.py
```

The benchmark seed, test files, schemas, and generated reference vectors are included. The external exact TOI datasets and third-party libraries are not redistributed; their project pages and papers are cited for the next validation gate.

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